



JAIN
DEEMED-TO-BE UNIVERSITY

SCHOOL OF
COMPUTER
SCIENCE AND IT

DEPARTMENT OF CS & IT

Programme: MCA

Activity-1
Cloud Analytics

25MCASC302

Name: G K Jeeva

USN: 25MCAR0158

Semester: 3rd Semester

Branch: MCA-SCT

Submitted to: Dr. Prabhakaran Abraham

Professor

School of CS & IT

DESIGN AND DEPLOYMENT OF A PERSONAL PORTFOLIO WEB APPLICATION

1. Introduction

This report documents the design, development and deployment of a personal professional portfolio website. The portfolio presents the student's profile, education, technical skills, projects, experience and contact details in one accessible web application. The project was created to meet the Activity 1 requirement of designing a portfolio, maintaining the complete project in GitHub and publishing a working live link. The solution is intentionally simple, responsive and easy for recruiters or academic evaluators to navigate.

The website is developed using HTML, CSS, and JavaScript and is designed to provide easy navigation across different sections such as Home, About, Skills, Projects, Education, and Contact. It also includes links to GitHub and LinkedIn and a contact form that allows visitors to send messages.

Project objectives

- To develop a professional and responsive personal portfolio website.
- To showcase academic qualifications, technical skills, and projects.
- To maintain the project source code using Git and GitHub.
- To deploy the portfolio on Microsoft Azure Static Web Apps.
- To implement automated deployment using GitHub Actions.
- To provide a contact form for communication with visitors.

Portfolio structure

The portfolio is designed as a single-page web application with a simple and user-friendly navigation structure. The website is divided into different sections to present personal and professional information in an organized manner.

The main sections of the portfolio are:

- **Home:** Provides a brief introduction and highlights the profile as a Cloud and DevOps enthusiast.
- **About:** Describes the academic background, specialization, and career interests.
- **Skills:** Presents technical skills related to cloud computing, DevOps, programming, web development, databases, and development tools.
- **Projects:** Showcases academic and practical projects related to AWS, Azure, Terraform, Docker, Kubernetes, and web development.
- **Education:** Displays academic qualifications, including MCA and BCA.
- **Contact:** Provides email, GitHub, and LinkedIn links along with a contact form for visitors to send messages.

2. Design and Development

The personal portfolio is designed as a single-page, responsive web application with a clean and professional interface. The design focuses on presenting personal information, technical skills, projects, education, and contact details in a structured and easy-to-navigate format.

Design Approach

- Simple and professional layout: Clear headings, sections, cards, buttons, and consistent spacing are used to improve readability.
- Responsive design: The website is designed to work on desktops, laptops, tablets, and mobile devices.
- Section-based navigation: Users can easily navigate between Home, About, Skills, Projects, Education, and Contact sections.
- Professional presentation: GitHub, LinkedIn, and email links are provided for professional communication.
- Interactive elements: Smooth scrolling, mobile navigation, active navigation highlighting, and scroll-based animations improve the user experience.

Development Approach

The portfolio is developed using HTML, CSS, and JavaScript. HTML is used to create the structure and content of the website, while CSS is used for styling, layout, responsiveness, and visual design. JavaScript is used to implement interactive features such as the mobile menu, smooth scrolling, active navigation, and reveal animations.

The project source code is organized into separate files such as index.html, css/style.css, and js/script.js, making the application easier to maintain and modify.

The completed project is stored in GitHub and deployed on Microsoft Azure Static Web Apps. GitHub Actions is used for automated deployment whenever changes are pushed to the main branch. The contact form is integrated with Form Submit, allowing visitors to send messages without requiring a separate backend server.

3. Technologies, Tools and Platforms

The portfolio project uses a combination of web development, version control, automation, and cloud technologies. Each technology is used for a specific purpose in developing and deploying the portfolio.

Area	Technology
Frontend	HTML5, CSS3, JavaScript
Styling	CSS
Client-side Interaction	Java Script
Source Code Repository	GitHub
CI/CD Automation	GitHub Auction
Cloud Platform	Azure Static Web Apps
Contact Form	Form Submit

4. Development and Implementation Approach

The portfolio was developed as a static web application using HTML, CSS, and JavaScript. The development approach focused on creating a simple, responsive, and professional website that can be easily maintained and updated. The main webpage is created using HTML, which provides the structure for sections such as Home, About, Skills, Projects, Education, and Contact. CSS is used to design the layout, colors, spacing, cards, buttons, and responsive behavior for different screen sizes.

JavaScript is used to add interactive functionality to the website. It provides the mobile navigation menu, smooth scrolling between sections, active navigation highlighting, and reveal animations when content enters the screen. The source code is maintained in GitHub, and GitHub Actions is used to automate the deployment process. The website is deployed on Microsoft Azure Static Web Apps, with the application hosted directly from the repository. The contact form is integrated with Form Submit, allowing visitors to send messages to the configured email address without requiring a separate backend server.

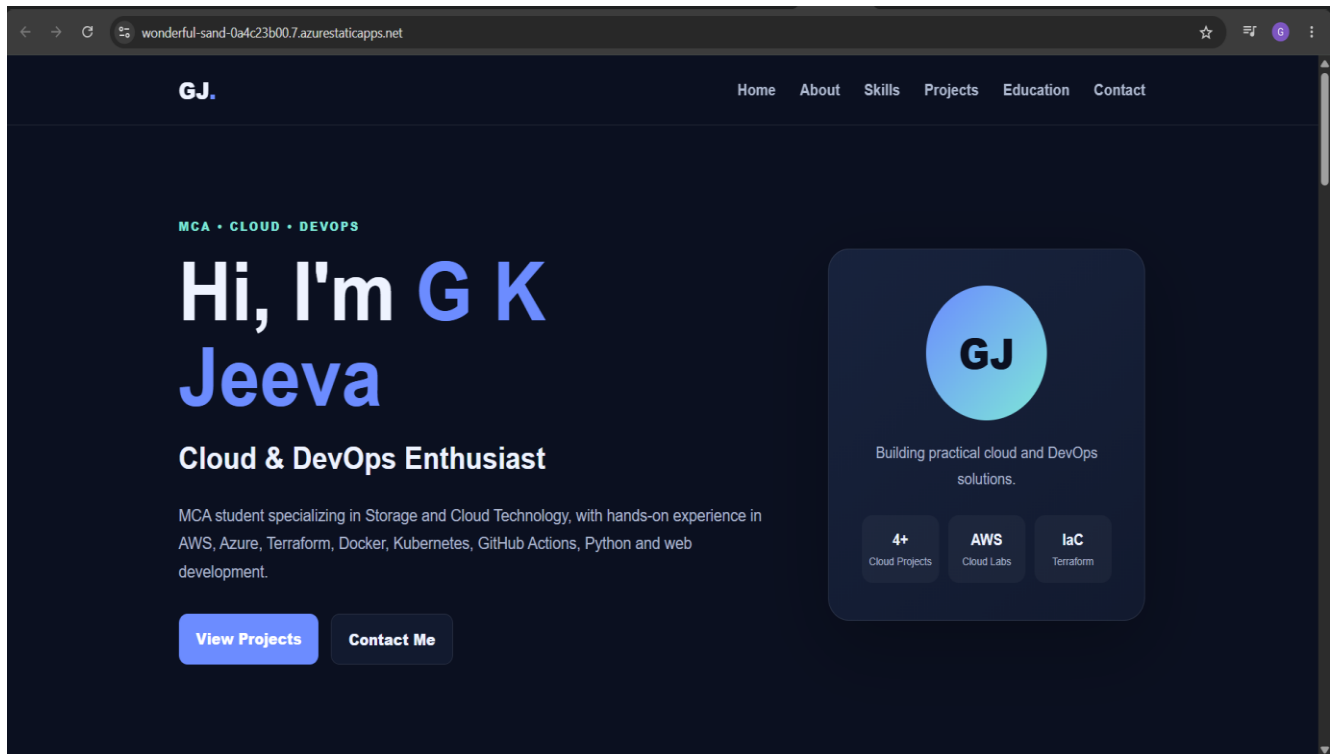
5. Challenges faced and solutions

During the development and deployment of the personal portfolio, several challenges were encountered. These challenges provided practical experience in troubleshooting web applications, GitHub, and Azure deployment.

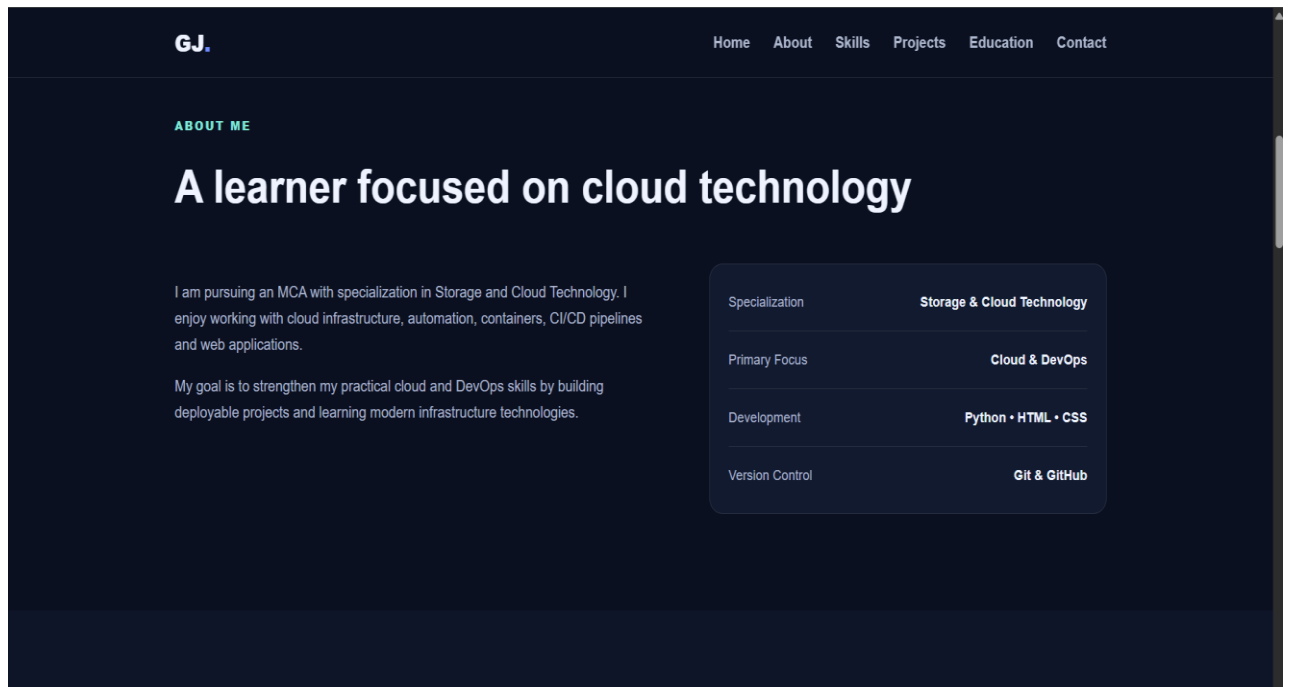
- **Contact Form Issue:** The form was not submitting correctly. It was fixed using FormSubmit.
- **Azure Deployment Issue:** Azure was showing an older version. GitHub Actions was configured to deploy the latest changes.
- **Deployment Token Issue:** GitHub Actions initially failed due to a missing Azure deployment token. The token was added as a GitHub secret.
- **Testing Issue:** The contact form was tested through the live Azure website instead of opening the HTML file locally.

6. Portfolio Screenshots: Overview

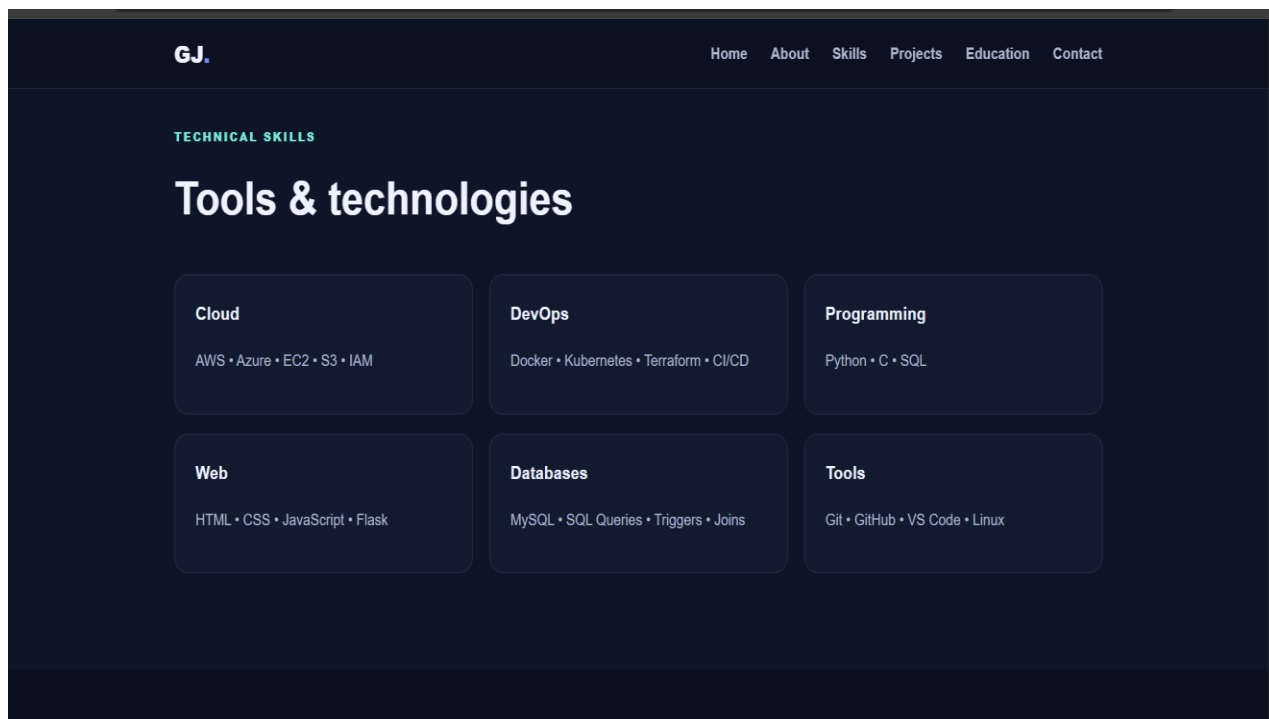
Portfolio home / landing page with introduction and navigation



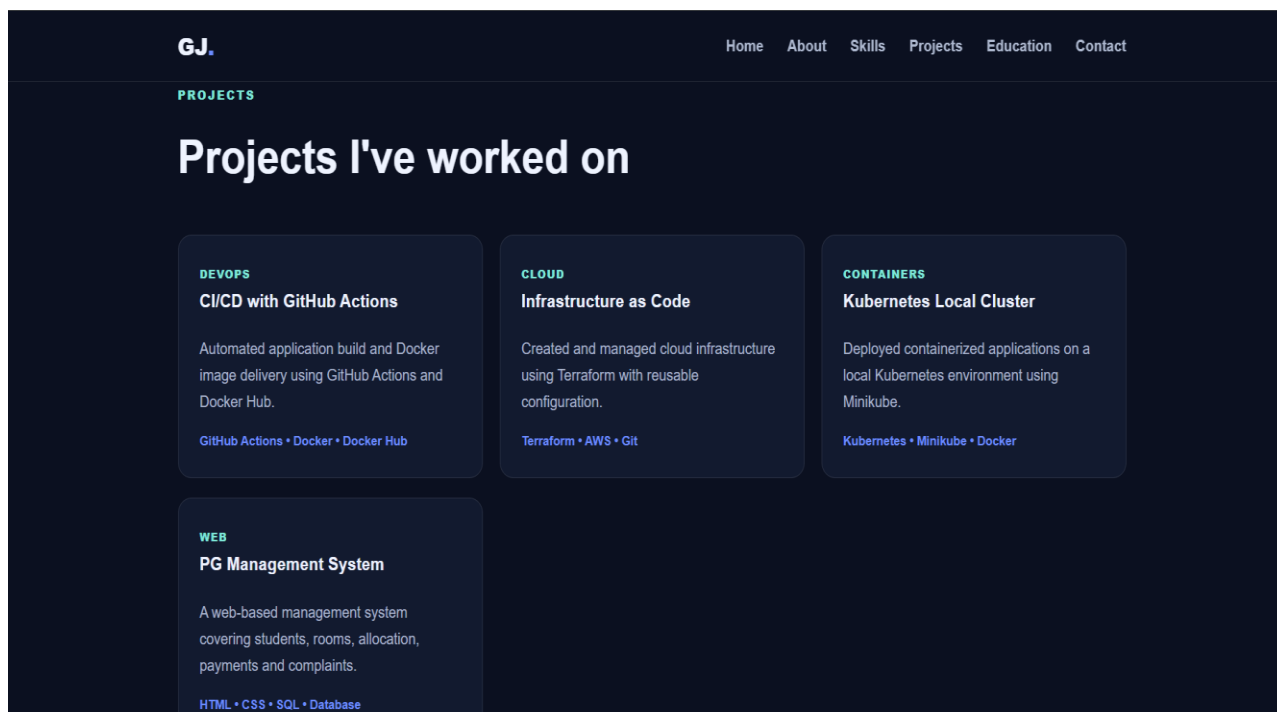
About



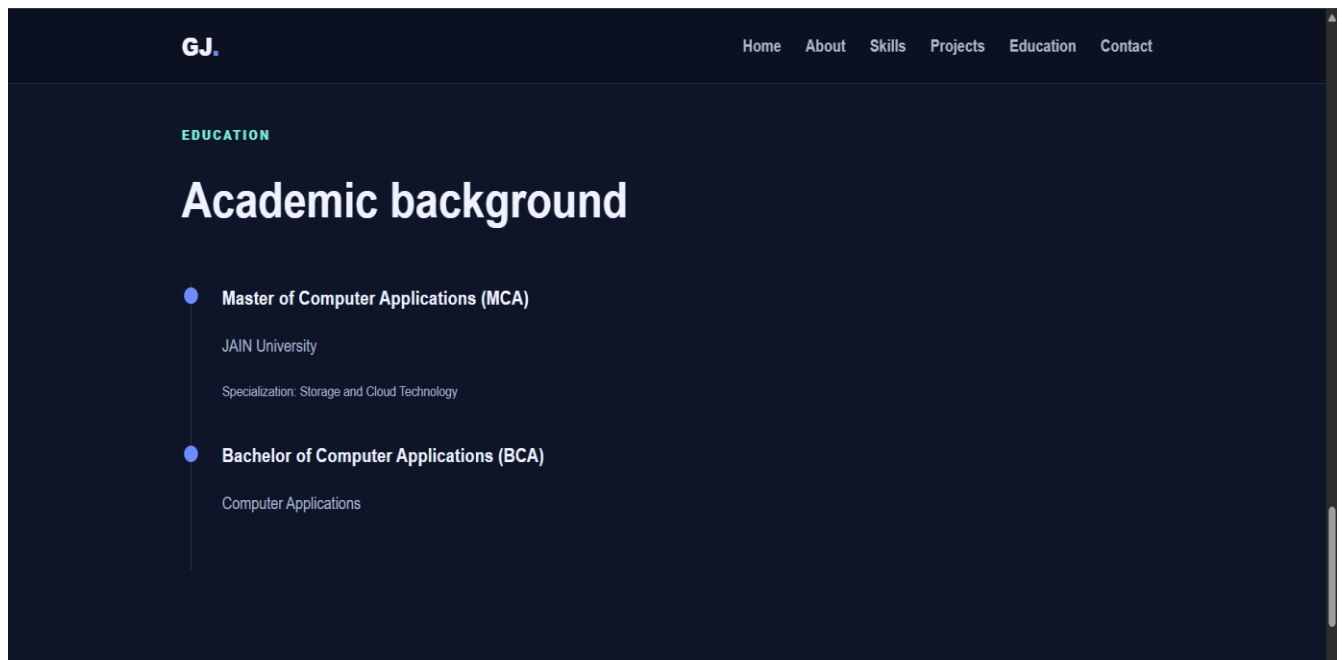
Skill section



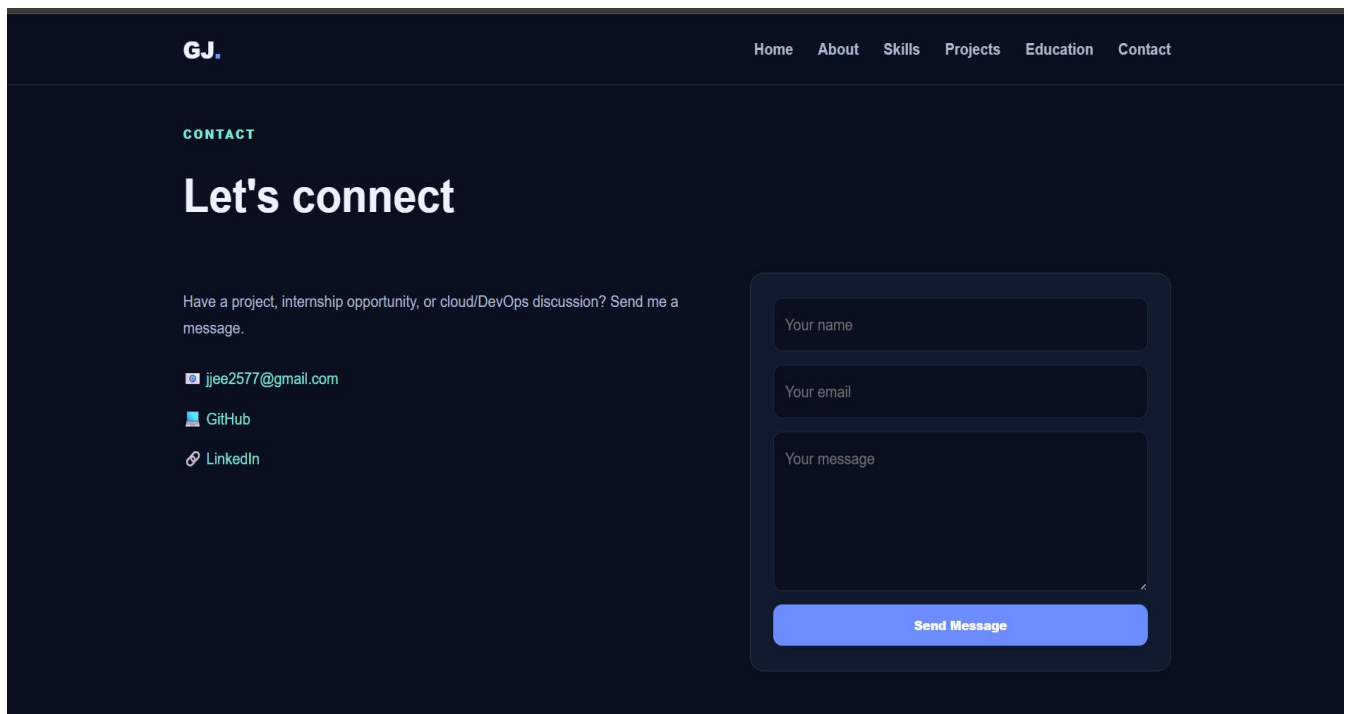
Project section



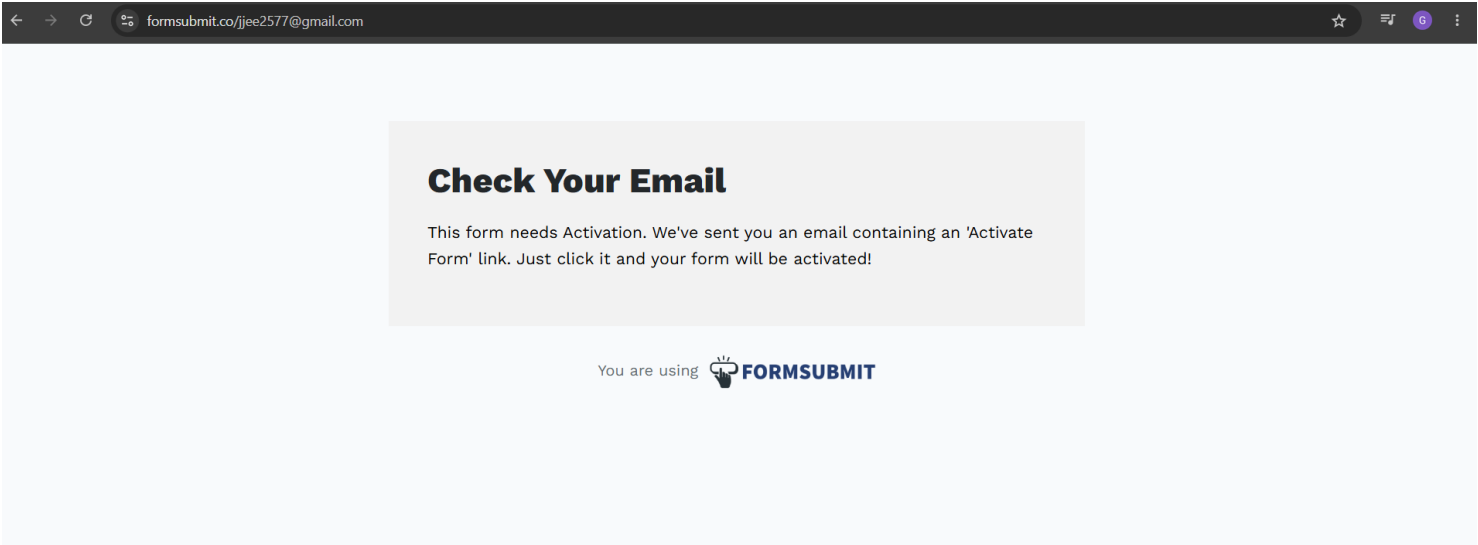
Education Section



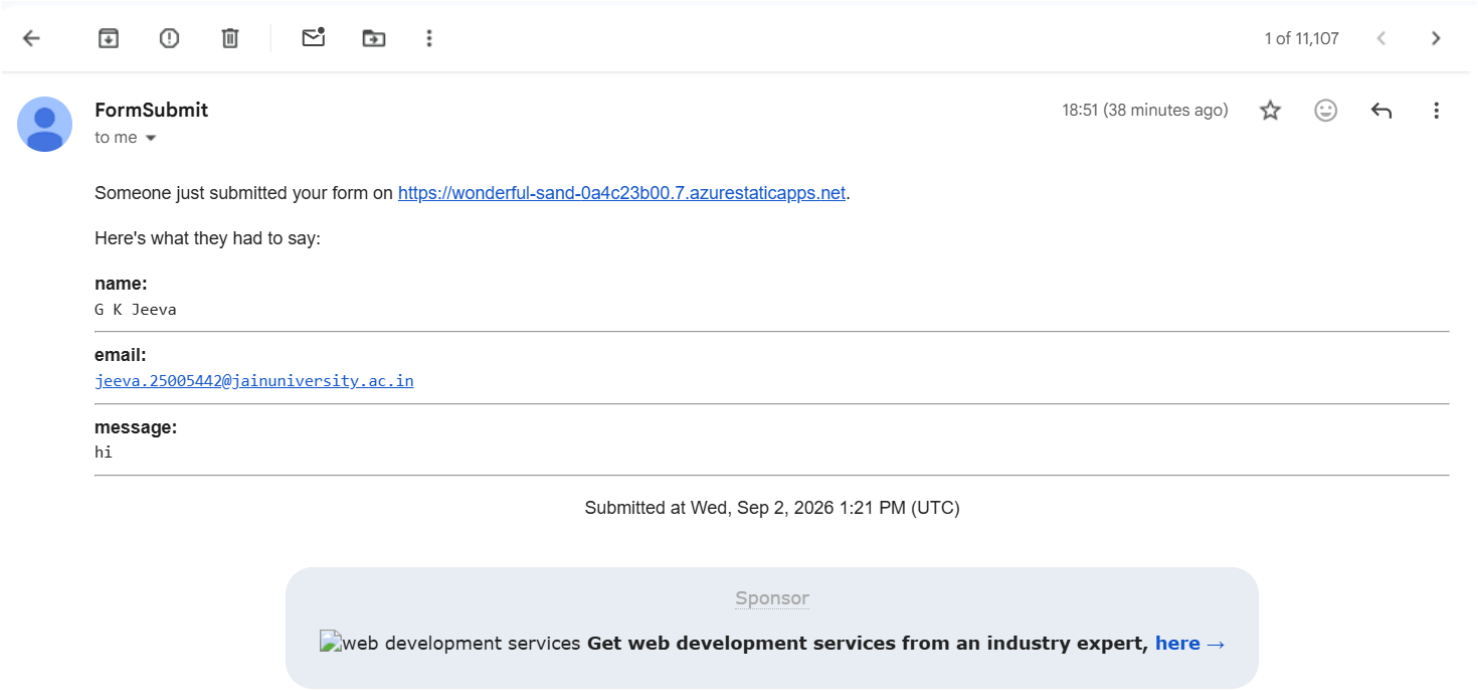
Contact Section



Form Submit



Response

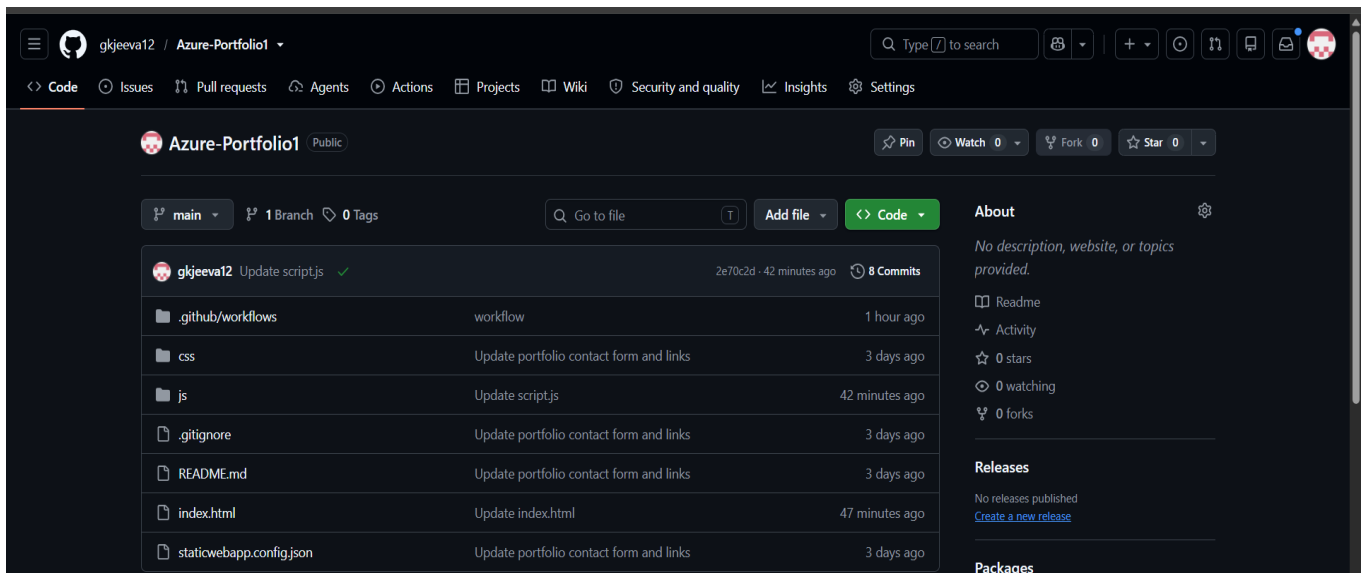


Successful GitHub Actions build and deployment workflow

7. GitHub Repository and CI/CD Evidence

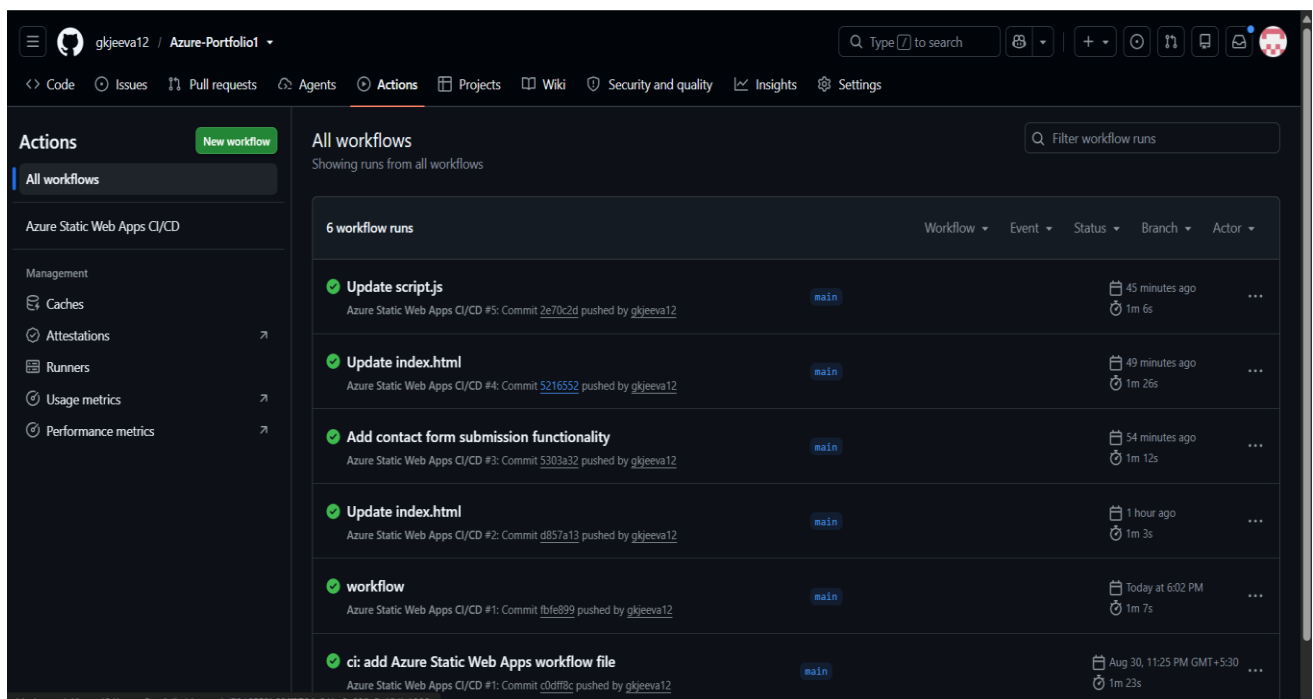
The complete portfolio project is maintained in the GitHub repository gkjeeva12/Azure-Portfolio1. GitHub is used for source code management and version control.

GitHub repository structure for the portfolio project



V

Workflow



8. Azure Deployment

The portfolio is deployed to Microsoft Azure Static App Service under the application name Jeeva-portfolio. Azure App Service provides the hosted environment for the live portfolio, while GitHub Actions provides the automated CI/CD path from repository updates to deployment.

This deployment approach separates development from hosting: source changes are managed in GitHub, the workflow runs the build process, and Azure serves the completed application. The successful workflow evidence confirms the configured pipeline completed its build and deploy stages.

Successful deployment of the resource

The screenshot displays the Microsoft Azure portal interface for a Static Web App named 'jeeva-portfolio'. The left sidebar contains navigation options such as Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Resource visualizer, Settings, Routes, Configuration, Snippets, Environment variables, Application Insights, Custom domains, APIs, and Environments. The main content area shows the 'Overview' tab with the following details:

- Essentials:**
 - Resource group (move): [RG-CA-core](#)
 - Subscription (move): [IAIN-CloudAnalytics-Lab](#)
 - Subscription ID: c2135661-f8fb-457d-abac-225af670f33e
 - Location: Global
 - Sku: Free
 - Tags (edit): [Add tags](#)
- URL:** <https://wonderful-sand-0a4c23b00.7.azurestaticapps.net>
- Source:** [main \(GitHub\)](#)
- Deployment history:** [GitHub Action runs](#)
- View workflow:** [azure-static-web-apps-wonderful-sand-0a4c23b00.yml](#)

Below the essentials, there are tabs for 'Get started', 'Monitoring', and 'Deployment history'. The 'View your application' section shows:

- Status:** Ready
- Environment:** Production
- Domain:** <https://wonderful-sand-0a4c23b00.7.azurestaticapps.net>
- Hosting plan:** Free

A 'Visit your site' button is available. At the bottom, a progress bar indicates 'Prepare for production (0/3 completed)'.

9. Conclusion

The personal portfolio project was successfully designed, developed, and deployed as a responsive web application using HTML, CSS, and JavaScript. The website provides a professional platform to showcase academic qualifications, technical skills, projects, education, and contact information. The project provided practical experience in web development, Git and GitHub, GitHub Actions, and Microsoft Azure Static Web Apps. The use of GitHub Actions helped automate the deployment process, making it easier to publish updates to the live website.

The contact form was integrated with Form Submit, allowing visitors to send messages without requiring a separate backend server. The project also involved troubleshooting deployment and contact-form issues, which improved practical understanding of cloud hosting and CI/CD workflows. Overall, this project helped strengthen practical knowledge of cloud computing, DevOps, version control, web development, CI/CD automation, and Azure deployment, while also creating a professional portfolio that can be used for academic and career purposes.

10. Submission Links

The completed portfolio project is maintained in the GitHub repository, which contains the source code, project configuration, build files and deployment workflow.

GitHub Repository:

<https://github.com/gkjeeva12/Azure-Portfolio1>

Live Portfolio:

<https://wonderful-sand-0a4c23b00.7.azurestaticapps.net>